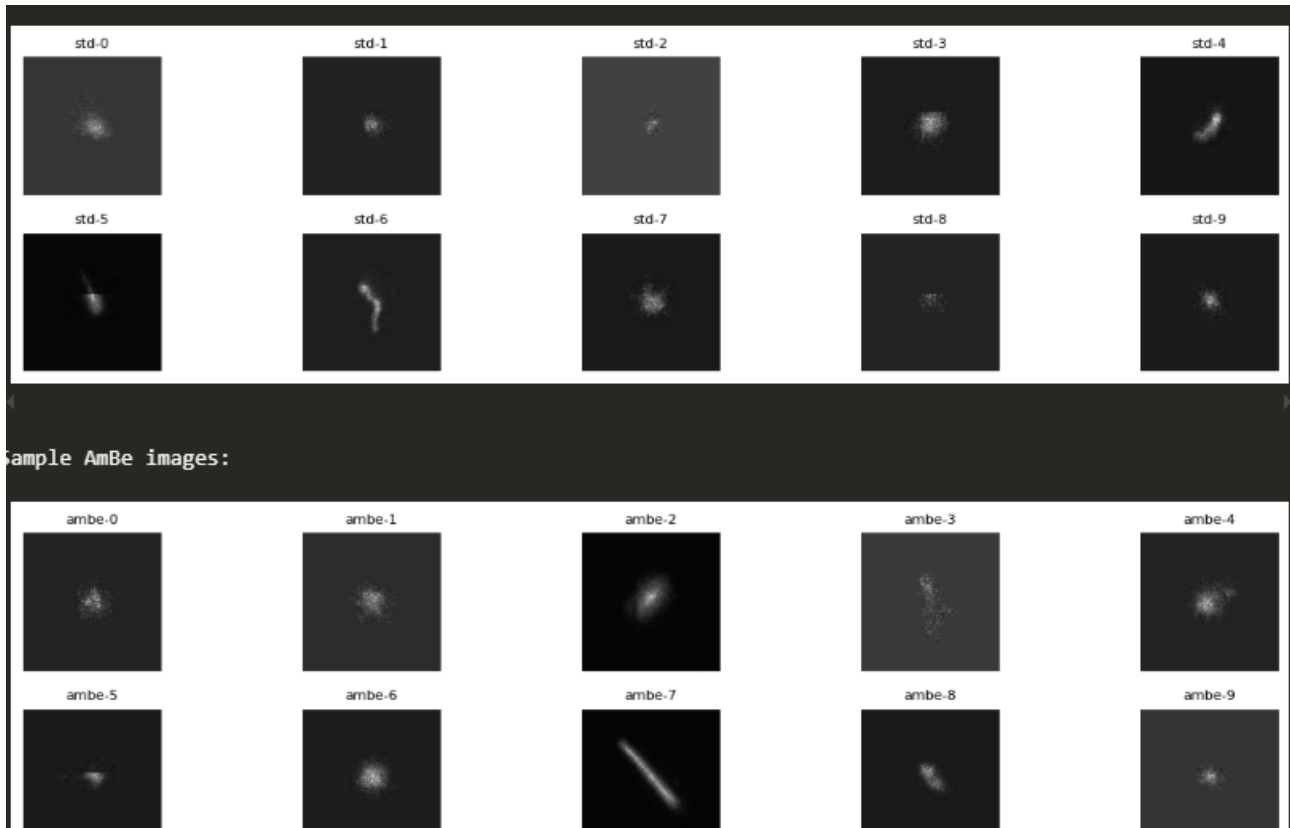


Starting Stuff (Common to All Parts)

- **UPLOAD:** Grid with extracted images (just as reference)



- **UPLOAD:** Brief Explanation of Electronic and Nuclear Recoils (morphology, not the process)

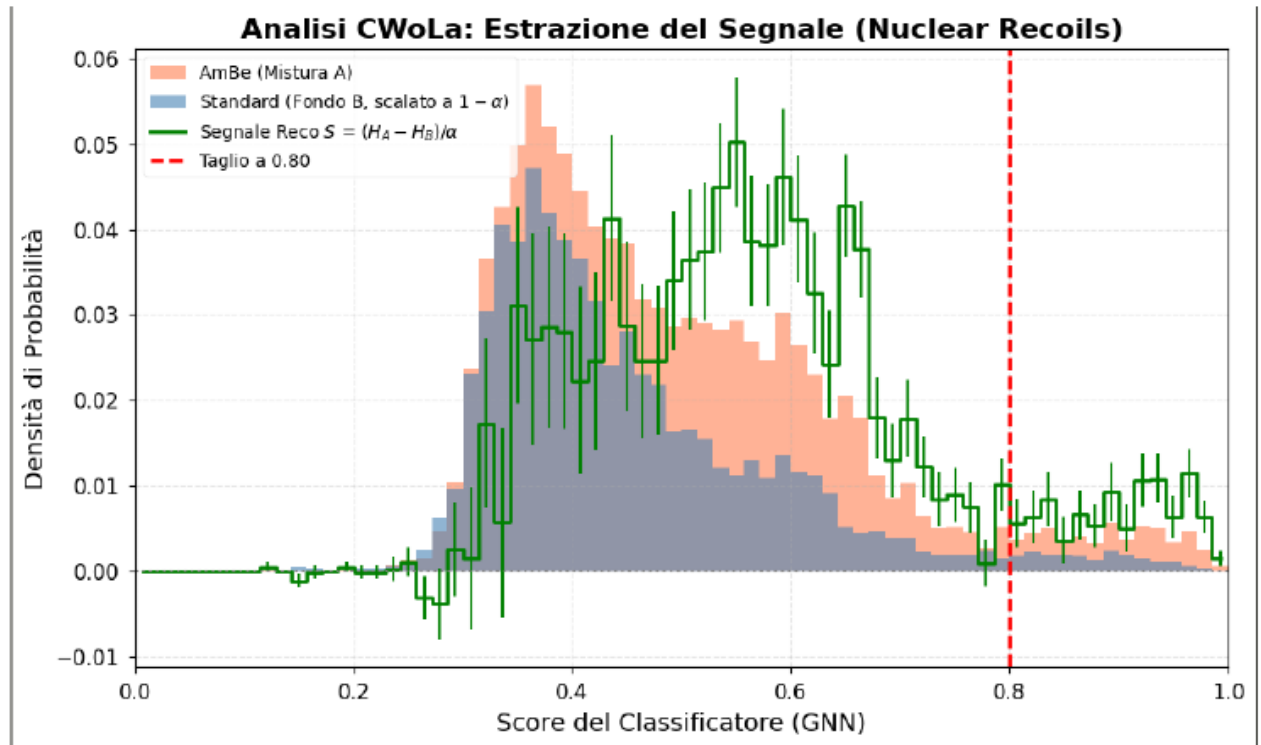
DIEGO

PART 1

- **UPLOAD:** ROC AUC and Loss

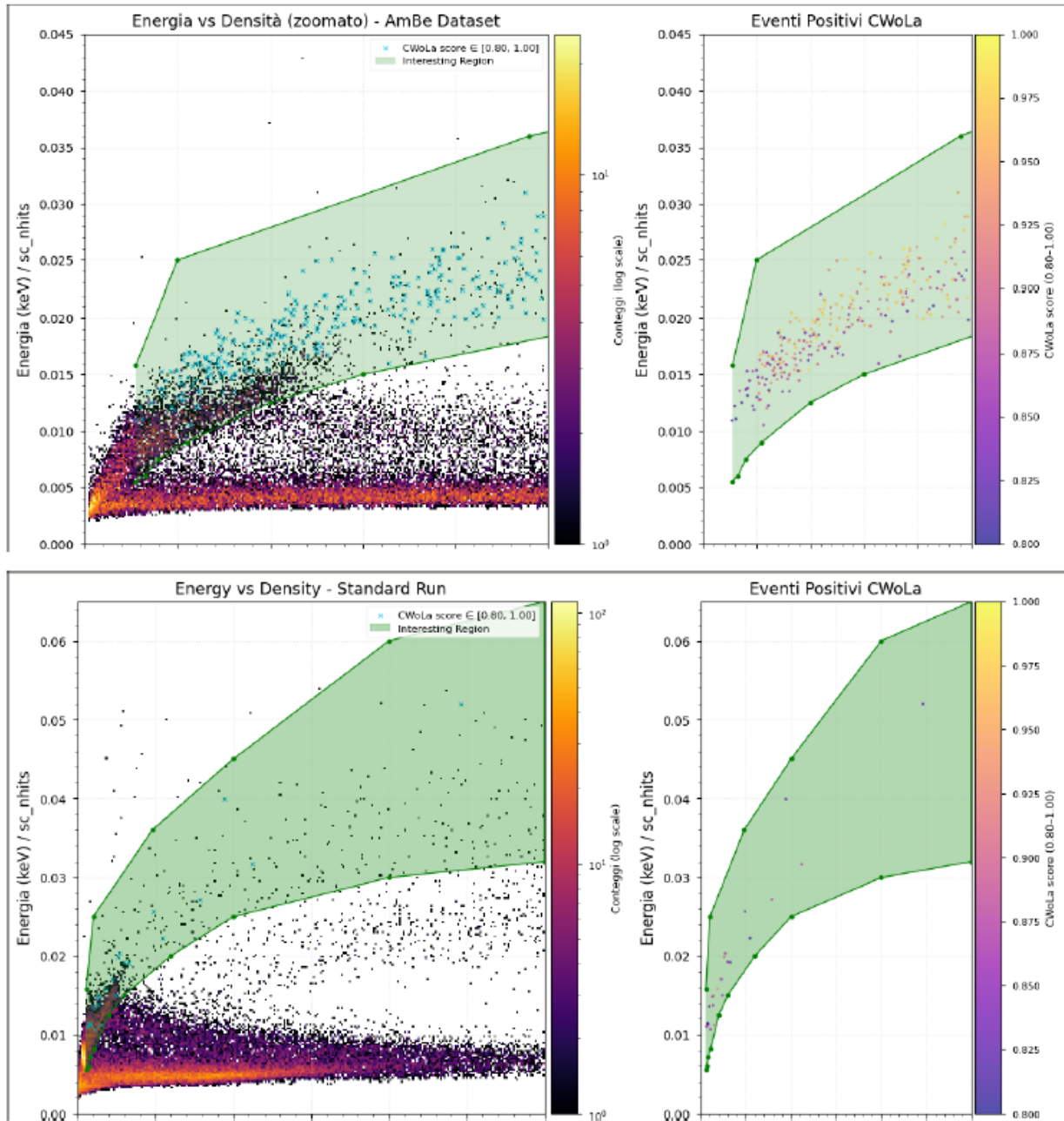
- **UPLOAD:** Score_Distribution exp1 MC_supervised (applied on Test Monte Carlo data) and Uncertainty Distribution + **MISSING:** reconstructed signal (check AML PROJECT.ipynb) + **MISSING:** normalization for STD dataset to 1-alpha (alpha=0.36) and AmBe normalization to 1 [take inspiration by the code in AML PROJECT.ipynb]

Representation of Score Distribution for the Mixed AmBe Real Dataset



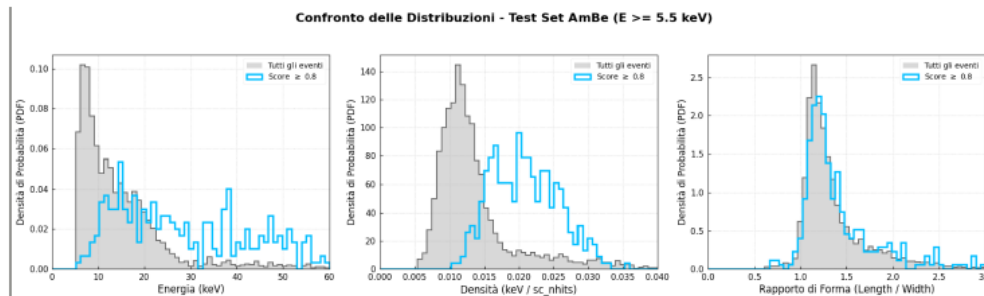
- **UPLOAD:** Mean Uncertainty vs Energy
- **EXPLANATION:** mean uncertainty sometimes is 0 for some points because since the model is trained on actually label data, it has a stronger starting point, is more confident, and therefore classifies more confidently
- **MISSING:** Application of the model (the one trained on Simulated MC) on Real Data to have an idea of performance (score distribution FULL (with reconstructed signal), uncertainty distribution)

- **MISSING:** distribution of events in the energy-density plane (with score as crosses, as in the image) ONLY FOR REAL DATA, CAUSE IT NEEDS THE PARQUET



- **MISSING:** score correlation with variables (on Real Data from Parquet files. They probably are not on the GitHub cause it wont let me upload files heavier than 25 MB, but if you prepare the code I'll run it for you)

Score correlation with variables



- **REMOVE:** AUC vs Energy graph (it's completely nonsense)
- **EXPLANATION:** we're not doing the Application of the GNN to the MC simulated data to perform the histograms for the various energies because since we trained the model on simulated data, the ones we actually apply the model to is a small fraction, so the histograms which get produced have too few images, making it not reliable as an analysis

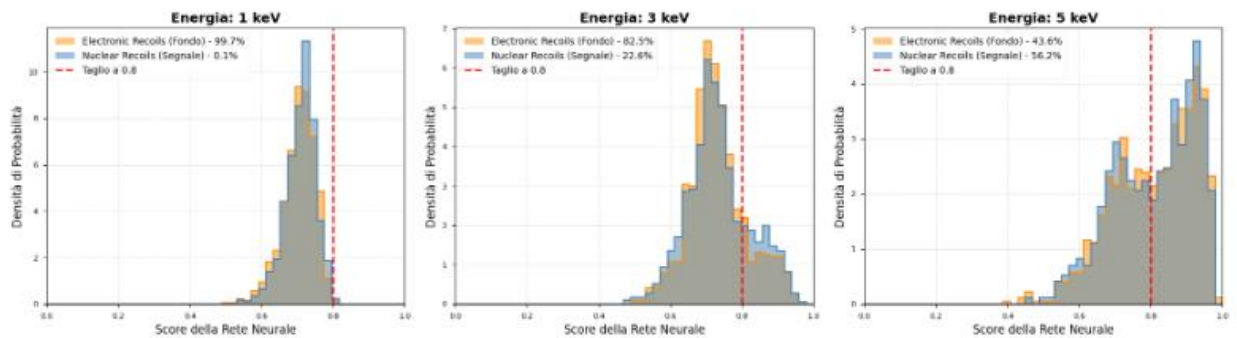
PART 2

- **REMOVE:** part 2 for the code (just focus on model trained on Simulated Data (label 1 and 0) and the Real CWoLa approach)

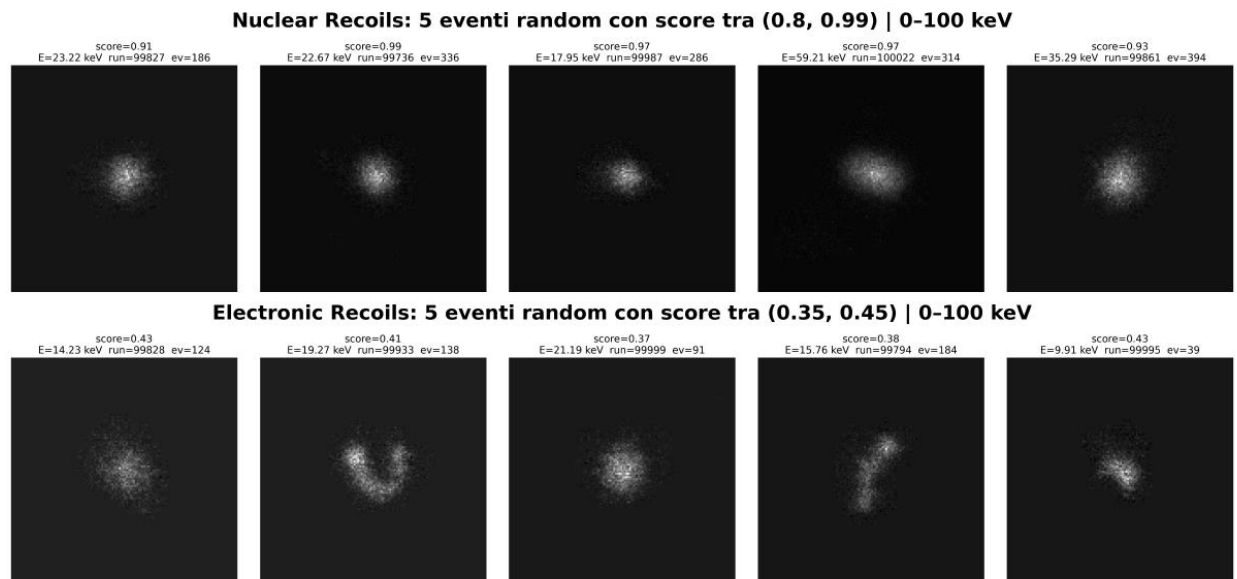
PART 3

- **UPLOAD:** ROC AUC and Loss (put a ylim (0.60-0.80) or smth like that to see it better)
- **UPLOAD:** Score_Distribution exp1 MC_supervised (applied on Test Monte Carlo data) and Uncertainty Distribution + **MISSING:** reconstructed signal + **MISSING:** normalization for STD dataset to 1-alpha (alpha=0.36) and AmBe normalization to 1
- **UPLOAD:** Mean Uncertainty vs Energy
- **EXPLANATION:** uncertainties are higher cause the model is trained using CWoLa, intrinsically more uncertain

- **MISSING:** Application of the model to the Monte Carlo data, divided by Energy, showing score distribution and uncertainty distribution

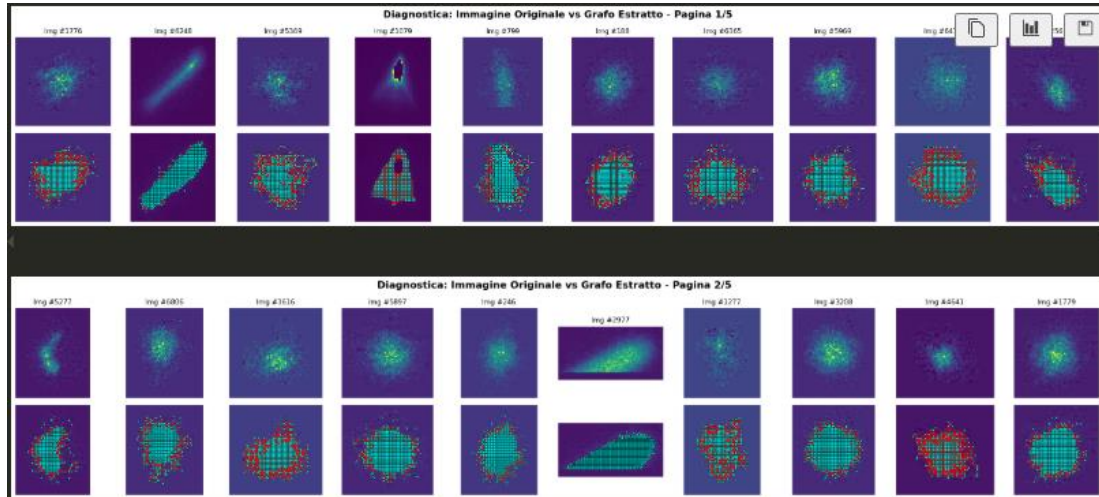


- **MISSING:** distribution of events in the energy-density plane (with score)
- **REMOVE:** Final Comparison
- **MISSING:** a graph where we show what images are classified with which score. Maybe showing some of the images with the highest uncertainty and the lowest uncertainty, to see why it is uncertain (justifying what the model is doing). I believe you will decide with criteria the right images to show hahaha

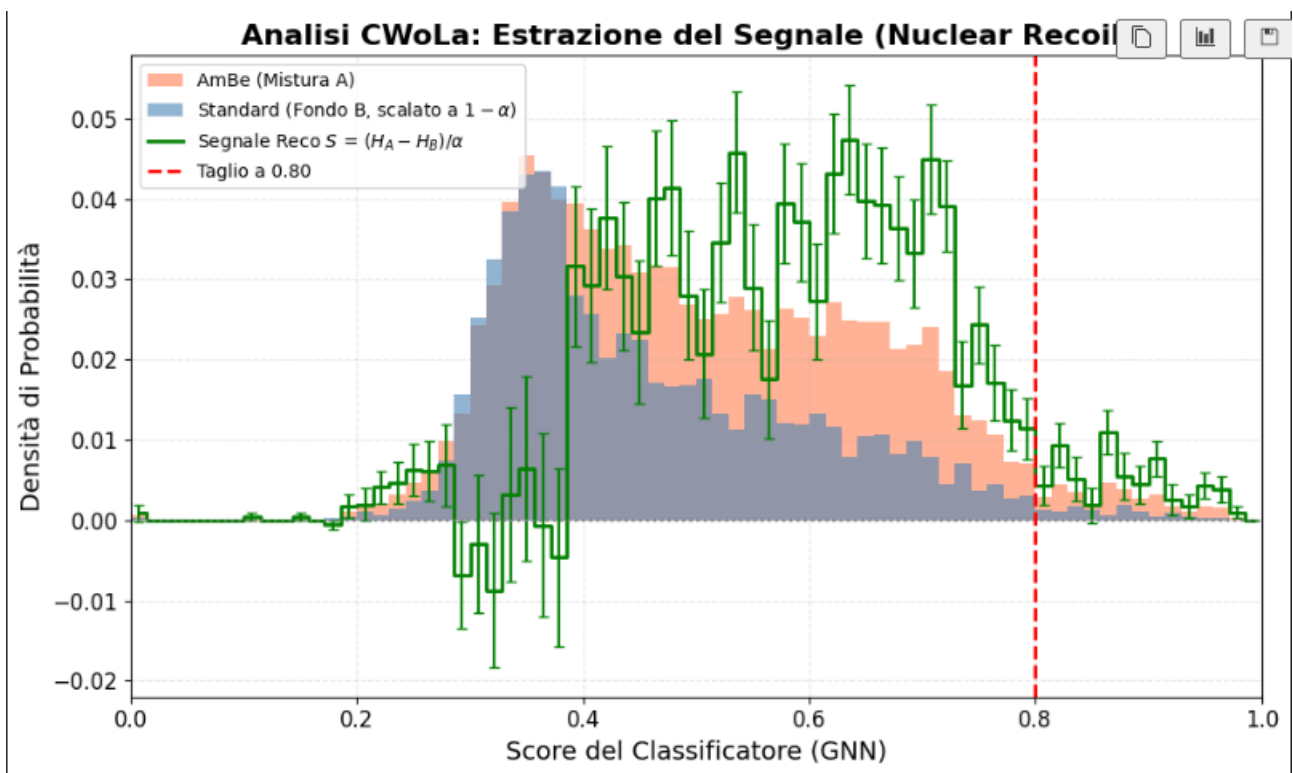


DAVIDE

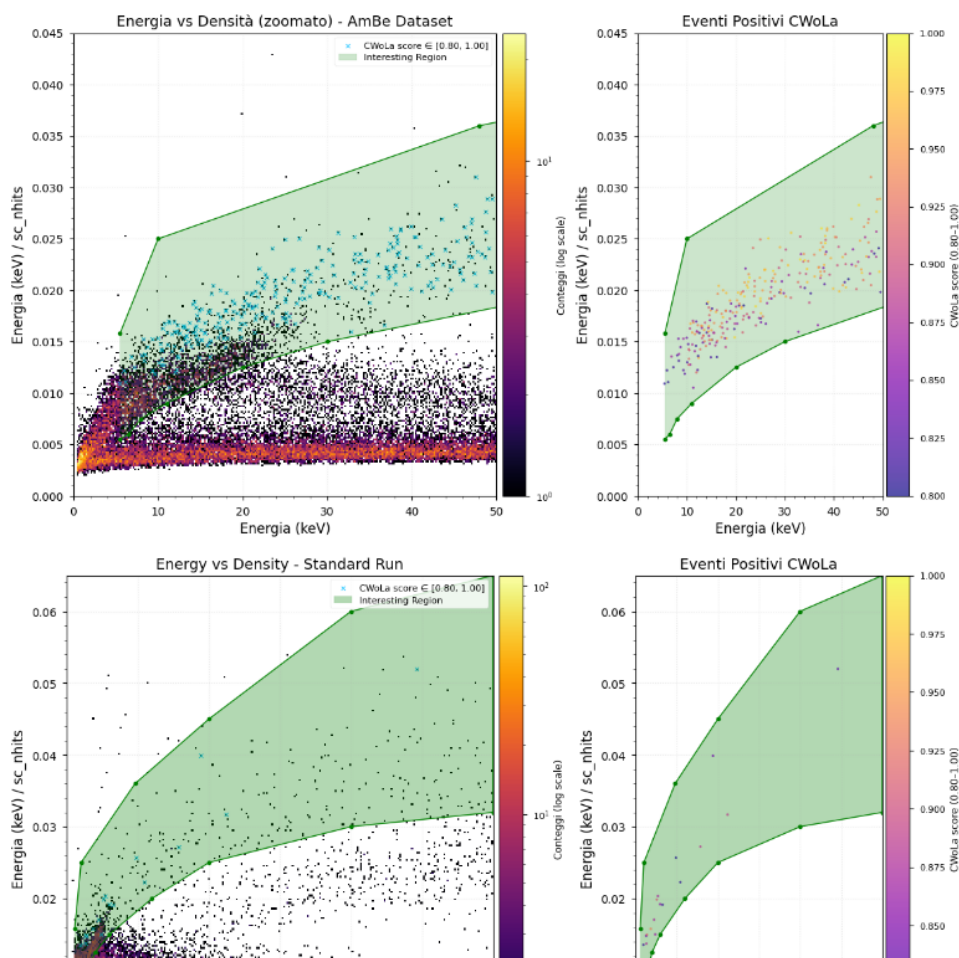
- **UPLOAD:** Extracted Graphs from Images + comparison (reduce number of graphs, just show 5 images per row)



- **UPLOAD:** Score Distribution (hoping it gets better...)

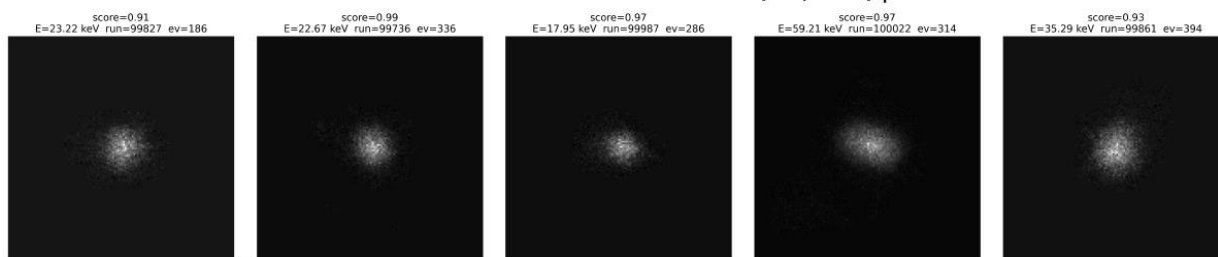


- **UPLOAD:** Score Distribution in the Energy Density Plane for both AmBe Dataset and STD Dataset

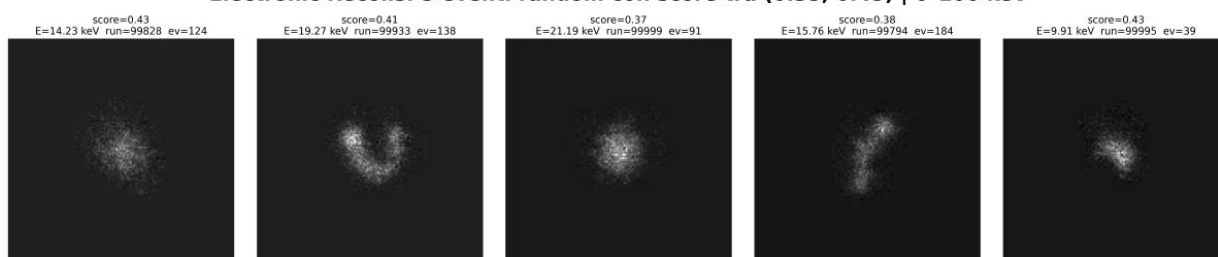


- **UPLOAD:** Correlation with the Reconstructed Variables
- **MISSING:** a graph where we show what images are classified with which score. As an example, setting a range of (0.8, 1.0) or (0.35, 0.45) or (0.45, 0.55), we should show what images the model is classifying, trying to see if the physical prediction adheres to the effective classification (we should add Energy and score as a thing shown in the image, as it is for the image shown below!! E=23.32 keV for example) – especially for the events classified with score 0.5

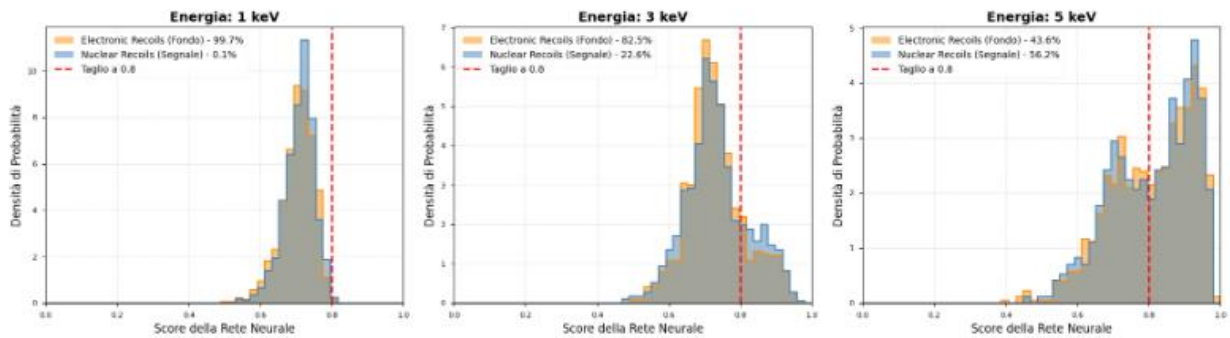
Nuclear Recoils: 5 eventi random con score tra (0.8, 0.99) | 0-100 keV



Electronic Recoils: 5 eventi random con score tra (0.35, 0.45) | 0-100 keV



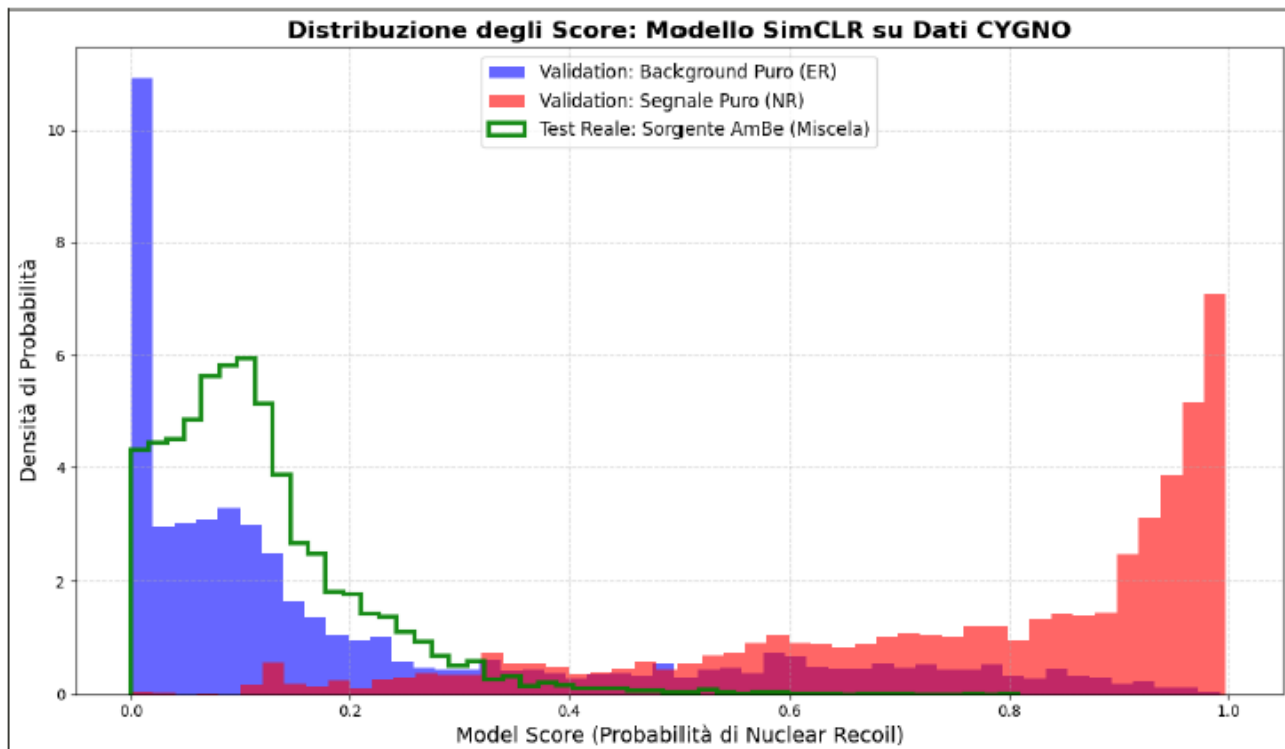
- **UPLOAD:** Distribution of Monte Carlo Events and Classification



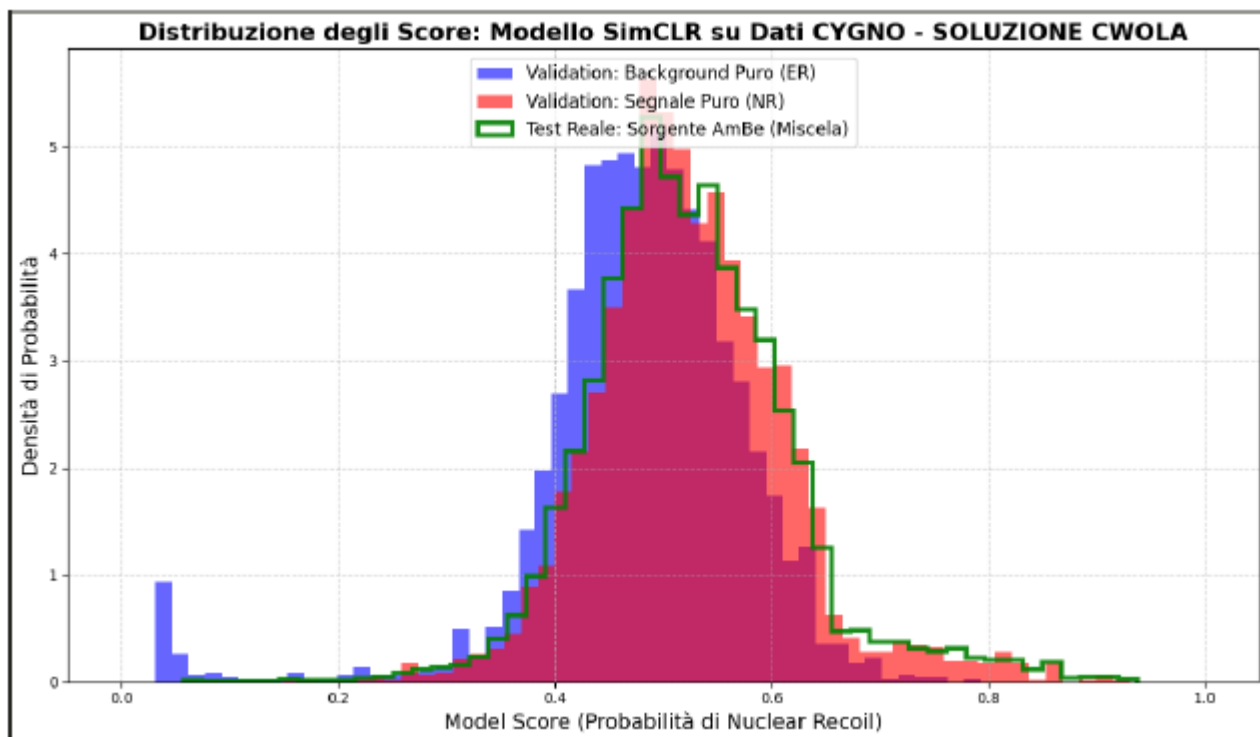
- **MISSING:** Code trained on Simulated Data + ALL Results:
 - Score distribution on Real AmBe data
 - Distribution in the Energy Density Plane
 - Correlation with Reconstructed Variables

MATTIA

- **EXPLAIN:** how does Contrastive Learning work, maybe add some images related to how we're expecting the Encoder to work (train on ALL data) and the way we're training the classifier
- **UPLOAD:** Initial Score Distribution for SIMCLR



- **UPLOAD:** Second Result: adding noise to NR
- **UPLOAD:** Third Result: applying CWoLa for Contrastive Learning



- **EXPLAIN:** why the results are so poor, what can we understand from these graphs
- **EXPLAIN/ MISSING:** what is the model really learning? How can we back-up our assumptions?
- **UPLOAD:** results with ENCODER TRAINED ON REAL DATA ONLY

- **EXPLANATION:** we're trying to make sure the model is learning NR vs ER, and not Real vs Simulated. In this way we're sure he cannot have an idea of what's simulated, cause he never sees it
- **EXPLAIN:** why is this approach is not working? the model is the best one, but its implementation is not that easy, we wanted to add features to the model for it to learn better but we had no time